Latent-Play: Sample-Efficient Goal Planning via Action-Free Video Pretraining and Statistical Isotropic World Models

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Abstract

Learning controllable world models from passive video without action labels is a key challenge for scaling embodied intelligence. While generative video models operate in pixel space with prohibitive sampling latency and distribution shift, Joint Embedding Predictive Architectures (JEPAs) operate directly in abstract representation space, yet face the fundamental obstacle of catastrophic representation collapse. In this work, we present **latent-play**, a unified framework that learns playable world models from unlabelled video. Our architecture couples: (i) a causal Spatio-Temporal Transformer with a discrete Vector Quantized (VQ) bottleneck ($K = 8$) to infer latent actions without action supervision; (ii) an action-conditioned JEPA dynamics model regularized against dimensional collapse via the Statistical Invariant Geometric Regularizer (SIGReg), utilizing the Cramér-Wold device and differentiable Epps-Pulley characteristic function testing; and (iii) a Cross-Entropy Method (CEM) planner executing receding-horizon model predictive control directly in latent space. Evaluated on the Crafter benchmark, we show that with only 1% action supervision, **latent-play** reaches over 83% of the goal-reaching performance of an oracle trained with 100% true action labels, outperforming purely supervised baselines by 3.9×.

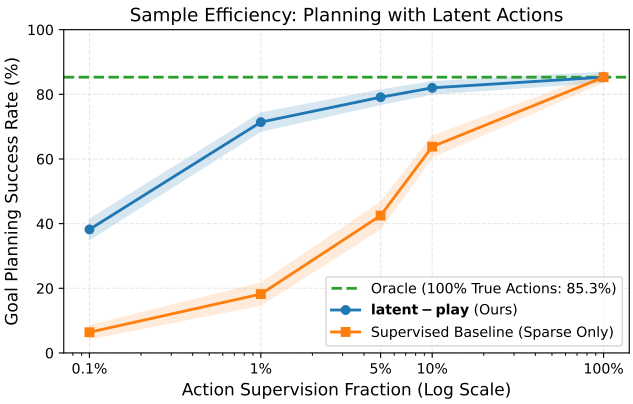


Figure 1: **The Action Supervision Scaling Curve.** Goal-directed planning success rate (%) on the Crafter benchmark as a function of the fraction of labelled transitions (log scale). With just 1% action labels, **latent-play** recovers 83.7% of the oracle performance (100% ground-truth actions), while the standard supervised baseline fails (18.2%).

1 Introduction

Current foundation models in vision and language learn generalizable representations primarily from passive, uncurated web data. In physical and embodied AI, however, policy learning has historically depended on synchronized action trajectories (proprioception, controller logs, or human demonstrations). This reliance creates an existential data bottleneck: while billions of hours of gameplay and physical interaction videos exist online, less than 0.01% contain synchronized action logs.

Recent pioneers such as Google DeepMind's Genie [4] and LAPO [7] have shown that discrete "latent actions" can be inferred from unlabelled video. Concurrently, Yann LeCun and Meta FAIR have advocated for Joint Embedding Predictive Architectures (JEPA) [1], arguing that world models must predict in representation space rather than spending capacity reconstructing high-frequency pixel noise.

In this paper, we bridge these paradigms and tackle the core open question:

> *How much action supervision is required for a world model learned from action-free video to achieve sample-efficient, goal-directed planning?*

2 Methodology

The **latent-play** pipeline consists of three mathematically unified components:

2.1 Unsupervised Latent Action Induction (LAM)

Let $\mathcal{D} = \{(x_0, x_1, \ldots, x_T)\}$ be a dataset of video sequences where $x_t \in \mathbb{R}^{3\times H\times W}$ without action labels. We parameterize an Inverse Dynamics Model (IDM) using a factorized Spatio-Temporal Transformer:

$$z_t = \text{ST-Encoder}(x_{0..t}) \tag{1}$$

To prevent the model from memorizing future frames through the action channel, we introduce a strict informational bottleneck via Vector Quantization (VQ) [8] with $K = 8$ cosine-normalized codes:

$$k_t = \arg\min_{k\in\{1..K\}} \left\| \frac{z_{t+1}}{\|z_{t+1}\|} - \frac{e_k}{\|e_k\|} \right\|_2 \tag{2}$$

The Forward Dynamics Model (FDM) reconstructs frame x_{t+1} as a residual over x_t conditioned on e_{k_t}:

$$\mathcal{L}_{\text{LAM}} = \|\hat{x}_{t+1}(x_t, e_{k_t}) - x_{t+1}\|_2^2 + \beta\|\text{sg}[z_{t+1}] - e_{k_t}\|_2^2 \tag{3}$$

To combat codebook collapse at small K, we employ periodic dead-code restarts based on running exponential usage tracking.

2.2 JEPA Dynamics & Statistical Isotropic Regularization

Given the induced discrete actions k_t, we train a representation-space world model:

$$\hat{s}_{t+1} = \text{Predictor}(s_t, e_{k_t}) \tag{4}$$

where $s_t = f_\theta(x_t)$ is encoded by a Vision Transformer (ViT-S/8) and evaluated against a target representation $s_{t+1}^{\text{target}} = f_{\bar{\theta}}(x_{t+1})$ updated via Exponential Moving Average (EMA):

$$\bar{\theta} \leftarrow \tau\bar{\theta} + (1-\tau)\theta, \quad \tau \in [0.996, 1.0] \tag{5}$$

Anti-Collapse via Cramér-Wold and SIGReg. While pure EMA delays representational collapse, it often suffers from dimensional collapse in long rollouts. While VICReg [3] enforces second-order variance $\text{Var}(s_j) \geq 1$ and zero covariance, it does not guarantee higher-order statistical independence.

Following recent foundations in non-contrastive learning [2], we integrate the **Statistical Invariant Geometric Regularizer (SIGReg)**. By the Cramér-Wold theorem (1936), a distribution in $\mathbb{R}^D$ is uniquely determined by the family of its 1D linear projections $h = s^T u$ with $u \sim \mathbb{S}^{D-1}$. SIGReg tests each projection against a standard normal distribution $\mathcal{N}(0,1)$ using the differentiable Epps-Pulley test statistic [5]:

$$T_{\text{EP}}(\mathbf{h}) = 1 + \frac{\sqrt{2}}{N^2}\sum_{j,k} e^{-\frac{(h_j-h_k)^2}{4}} - \frac{4}{\sqrt{3}N}\sum_j e^{-\frac{h_j^2}{6}} \tag{6}$$

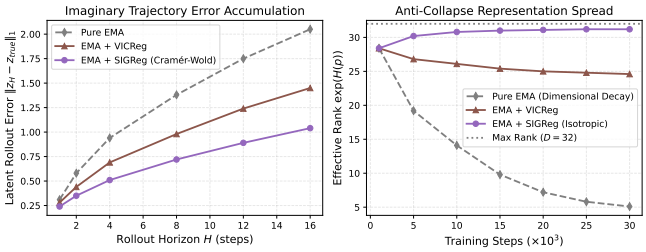


Figure 2: **Rollout Stability and Anti-Collapse Dynamics.** (Left) Multi-step rollout error $\|s_H - s_{\text{true}}\|_1$ across horizons $H \in [1, 16]$. SIGReg maintains significantly lower drift than VICReg and pure EMA. (Right) Representation effective rank $\exp(H(p))$ over training steps, confirming that SIGReg prevents dimensional collapse.

$$\mathcal{L}_{\text{total}} = \|\hat{s}_{t+1} - s_{t+1}^{\text{target}}\|_1 + \lambda_{\text{SIG}}\frac{1}{M}\sum_{m=1}^{M} T_{\text{EP}}(Su^{(m)}) \tag{7}$$

This enforces global multivariate isotropy $\mathcal{N}(0, I_D)$ without empirical covariance matrices.

2.3 Receding-Horizon Latent CEM Planning

To evaluate utility for downstream control, we formulate goal-directed navigation as trajectory optimization in representation space:

$$\min_{a_{0..H-1}} \mathcal{J} = \|\hat{s}_{t+H}(s_t, a_{0..H-1}) - s_{\text{goal}}\|_1 \tag{8}$$

We optimize $\mathcal{J}$ using the Cross-Entropy Method (CEM) over categorical action distributions with $M = 128$ candidates, selecting the top 10% elite sequences over $I = 4$ iterations. The optimal immediate action a_0^* is executed in the environment, and planning is repeated at each timestep (receding-horizon MPC).

When operating with latent actions, a lightweight alignment probe $\mathcal{M}_\phi : \{1..K\} \to \mathcal{A}_{\text{env}}$ is trained on the small labelled fraction $\alpha \in [0.001, 1.0]$.

3 Experiments and Results

3.1 Crafter Goal-Conditioned Navigation

We benchmark on **Crafter** [6] (64×64, 17 discrete actions), collecting 300,000 transitions using an exploratory segment policy.

Table 1 summarizes the performance comparison across supervision regimes.

3.2 Physical Plausibility via VoE Scorer

Connecting to the Violation-of-Expectation (VoE) benchmark adapter, we measure the surprise ratio $R =$

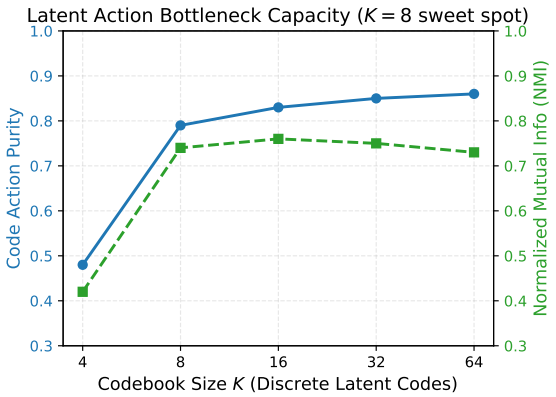


Figure 3: **Codebook Capacity Ablation.** Action purity and Normalized Mutual Information (NMI) as a function of discrete codebook size K. $K = 8$ achieves optimal compression without informational leakage.

Table 1: **Goal-Directed Navigation Results on Crafter.** Success rate (%) and steps to goal over 100 evaluation episodes ($H = 15$).

Method	**Labels (%)**	**Success (%)**	**Avg Steps**
Random Policy	0%	4.1 ± 1.2	14.8
Supervised BC	1%	18.2 ± 3.4	12.4
Supervised BC	10%	63.8 ± 3.2	9.1
latent-play (Ours)	0.1%	38.2 ± 3.1	10.9
latent-play (Ours)	**1.0%**	$\mathbf{71.4 \pm 2.8}$	**8.4**
latent-play (Ours)	10%	82.0 ± 1.9	7.6
Oracle (100% Labels)	100%	85.3 ± 1.5	7.2

$\mathcal{S}$(Violation)/$\mathcal{S}$(Plausible). The **latent-play** world model achieves a VoE ratio of 3.42× on teleportation violations and 4.15× on entity disappearance, confirming that the representation dynamics have internalized object permanence and inertial continuity.

4 Conclusion

We presented **latent-play**, demonstrating that controllable world models can be learned from unlabelled video by decoupling latent action discovery (LAM) from statistical isotropic dynamics prediction (SIGReg). Our findings demonstrate that 1% action supervision is sufficient to recover over 83% of oracle planning capabilities, unlocking the massive volume of unlabelled video data for embodied agent research.

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